

Sameer Alam

Associate Professor | Director, Air Traffic Management Research Institute (ATMRI) | Nanyang Technological University, Singapore

Google Scholar: scholar.google.com/citations?user=5W6FyV0AAAAJ

Professional Profile

Academic leader and researcher with more than two decades of experience in AI-enabled air transportation systems, air traffic management, complex systems, human-AI collaboration and sustainable aviation. Current Director of the Air Traffic Management Research Institute (ATMRI) at Nanyang Technological University, Singapore, with a research record spanning foundational modelling, operational deployment, international regulation and policy engagement.

83	150+	4,569	38
Journal papers	Conference papers	Google Scholar citations	Google Scholar h-index
S\$28M+	S\$13.46M	11	20+ yrs
External research funding	Future Airspace Architecture	Best paper awards	Academic / industry experience

Academic Qualifications

2008	PhD (Computer Sc.), University of New South Wales (UNSW), Australia
1999	Master’s degree (Computer Sc.), Birla Institute of Technology, Mesra, India
1996	Master’s degree (Economics), University of Rajasthan, India
1994	Bachelor’s degree (Mathematics), University of Rajasthan, India

Academic & Professional Appointments

Jan 2025 - Present	Director, Air Traffic Management Research Institute*, NTU Singapore *ATMRI is a pan-university research institute jointly established by CAAS and NTU
Nov 2019 – Dec 2024	Deputy Director, Air Traffic Management Research Institute, School of Mechanical & Aerospace Engineering, NTU Singapore
Sep 2019 – Apr 2024	Co-Director, SAAB-NTU Joint Research Lab, School of Mechanical & Aerospace Engineering, NTU Singapore
Jul 2017 - Present	Associate Professor, School of Mechanical & Aerospace Engineering, NTU Singapore
Jul 2017 – Mar 2024	Visiting Fellow, School of Engineering & Information Technology, University of New South Wales (UNSW), Canberra, Australia
Jul 2016 - Jul 2017	University Senior Lecture (Aviation), School of Engineering & Information Technology, University of New South Wales (UNSW), Canberra, Australia
Jul 2016 - Jan 2017	Visiting Scientist, Ecole Nationale de l'Aviation Civil (ENAC), Toulouse, France
Jan 2013 - Jun 2016	University Lecturer (Aviation), School of Engineering & Information Technology, University of New South Wales (UNSW), Canberra, Australia
Jan 2011 - Dec 2013	Senior Research Associate (senior Post Doc), School of Engineering & Information Technology, University of New South Wales (UNSW), Canberra, Australia
Jan 2008 - Dec 2011	Research Associate (Post Doc), Defence & Security Application Research Centre, University of New South Wales (UNSW), Canberra, Australia
Feb 2007 - Dec 2009	Research Assistant, School of IT & Electrical Engineering, University of New South Wales (UNSW), Canberra, Australia

Honours & Awards

Year	Academic Honour / Award
2011	Young Tall Poppy Scientist of the Year 2011 Award (Australian National Competition for Early Career Researchers), Australian Institute of Policy & Science, Australia
2009	NSi Inventor of the Year finalist, New South Innovation, UNSW, Sydney, Australia.
2008	Fresh Science Award (National competition and training program helping early-career researchers), Science in Public, Australia

Research Profile

Key Areas of Research

Air Traffic Management (ATM) – Advanced concepts and operational frameworks for next-generation airspace management, including trajectory-based operations, air traffic flow management, and human-AI collaboration for ATC decision support.

Artificial Intelligence (AI) – Development and application of AI/ML methods for predictive modelling, decision support, and intelligent automation in complex operational environments.

Complex Systems & Network Science – Modelling and analysis of large-scale, interconnected systems such as air traffic networks, including resilience, uncertainty management, and emergent behaviour dynamics.

Human-Machine Teaming– Cognitive engineering, controller support tools, and human-in-the-loop simulation.

Sustainability & Green Aviation – Modelling and optimisation to reduce emissions and environmental impact in ATM operations.

Research Funding

Role	Year	Project Title	Amount (S\$)	Source of Grant
Lead PI	2024-2027	Future Airspace Architecture Fostering Human-AI Hybrid Automation to cater to Future Air Traffic Demand.	13,457,704	Civil Aviation Authority of Singapore (CAAS)
Lead PI (NTU Singapore, CRIDA Spain, Eurocontrol FR, ENAC FR, LFV Sweden, Ingenuity i/o FR)	2024 - 2027	DeepFlow: AI enabled Dynamic Air Traffic Flow Configuration for Flow-Centric Airspace Management	0 (€1,388,764.50) (NTU do not receive cash funding, CAAS supplements the funding under NRF projects)	HORIZON-SESAR-2023-DES-ER-02 (Digital European Sky Exploratory Research)
Lead PI	2024-2025	A Top-Of-Descent Prediction Model for Optimal Continuous Descent Operation and its Integration in the AIR-LAB Ecosystem	435,305	Civil Aviation Authority of Singapore (CAAS)
Lead PI	2023-2028	ICAO and ASEAN Reports, and Regional ATM Studies	3,600,000	Civil Aviation Authority of Singapore (CAAS)-NTU
Lead PI (NTU Singapore, Uni Warwick, Eurocontrol FR, ENAC FR, LFV Sweden, NeoMetSys FR)	2023-2026	HyperSolver: Artificial Intelligence controller able to manage Air traffic Control (ATC) and Air Traffic Flow Management (ATFM) within a single framework	0 (€982,746.25) (NTU do not receive cash funding, CAAS supplements the funding under NRF projects)	HORIZON-SESAR-2023-DES-ER-01 (Digital European Sky Exploratory Research)
Lead PI	2023-2026	Artificial Intelligence for Air Traffic Management	381,875	A*STAR under First Executive Program of Scientific and Technological

Role	Year	Project Title	Amount (S\$)	Source of Grant
				Cooperation Italy and Singapore.
Lead PI	2022-2024	Mixed Airspace Integration CONOPS Models for Singapore	1,976,910	Economic Development Board (EDB)
Lead PI	2020 - 2023	ATP1 Air Traffic Conflict Prediction and Resolution	2,006,880	Civil Aviation Authority of Singapore (CAAS)-NTU
Lead PI	2020 - 2023	ATP2 Data-Driven Airspace and Airport Optimization	1,567,680	Civil Aviation Authority of Singapore (CAAS)-NTU
Lead PI	2020 - 2023	ATP3 Trust between AI-ATCO	1,765,560	Civil Aviation Authority of Singapore (CAAS)-NTU
Lead PI	2018 - 2024	Machine Learning for Airport Management and Tower Control	2,949,960	SAAB AB-NTU
Lead PI	2017 - 2018	Artificial Intelligence to Reduce Workload of ATCo-An Exploratory Study	230,400	Civil Aviation Authority of Singapore (CAAS)

Internal / Seed Funding

Role	Year	Project Title	Amount (S\$)	Source of Grant
PI	2024 - 2027	Research Project Incentive Scheme	10,000	Research Project Incentive Scheme
Lead PI	2023-2024	France-Singapore Workshop on Artificial Intelligence for Air Traffic Management	12,769.90	MERLION Programme 2023, NTU
PI	2022 - 2023	Managing Multiple Mental Models of Air Traffic Controllers in Flow-Centric ATC Using Intelligent Interactive Devices	100,000	NTU Internal
PI	2019 - 2020	Artificial Intelligence for Sustainable Air Transportation	12,000	COE Technology Primer Grant
PI	2018 - 2022	A Stochastic Runway-taxiway Simulator for Airfield Capacity Modelling	200,000	Start Up Grant, MAE, NTU

Citation Metrics

Database	Citation Count (with self-citations)	H-index
Scopus	2895	30
Google Scholar	4569	38

Research Awards & Recognition

Year	Research Awards
2025	Best Paper Award, 1 st FAA & Eurocontrol ARTD Symposium, Prague, Czech Republic
2023	Best Paper Award, 15 th US & Europe ATM R&D Seminar, Savannah, Georgia, USA.
2022	Best Paper Award, 10 th International Conference on Research in Air Transportation, Tampa, Florida, USA
2022	Best Paper Award, 41st AIAA/IEEE Digital Avionics Systems Conference (DASC), Portsmouth, VA USA
2019	Best Paper Award, 49th International Conference on Computers and Industrial Engineering, Beijing, China
2019	Best Paper Award, 13 th US & Europe ATM R&D Seminar, Vienna, Austria

Year	Research Awards
2018	Best Paper Award, 8 th International Conference on Research in Air Transportation, Barcelona, Spain
2015	Best Paper Award, 11th US-Europe ATM R&D Seminar, Lisbon, Portugal
2014	Best Paper Award, 6th International Conference on Research in Air Transportation, Istanbul, Turkey
2011	Best Paper Award, 9th US-Europe ATM R&D Seminar, Berlin, Germany
2010	Best Paper Award, 29th AIAA Digital Avionics Systems Conference, Utah, USA

Keynotes, Invited Presentations & Panels

Keynote Speaker, “*Artificial Intelligence in Aviation Safety*”, Workshop on Artificial Intelligence for Air Transportation (AI4AT), at the 40th AAAI Conference on Artificial Intelligence (AAAI-26), 26th Jan 2026, Singapore <https://ai4at.github.io/>

Keynote Speaker, “*Aviation Innovation with Artificial Intelligence*”, International Conference on Innovations in Computing & Applications (ICICA-2024), Science and Engineering Research Board of India, Birla Institute of Technology Mesra, Ranchi, Jaipur campus, 23rd May 2024, India, (Online) https://bitmesra.ac.in/UploadedDocuments/jpradmincse/NewsUPloaded/Newsca30ca0b49f4600babcbcb0643715fcb_ICICA2024briefreport.pdf

Keynote Speaker, “*Artificial Intelligence in Air Traffic Management*”, 39th International Federation of Air Traffic Controllers' Associations, Asia Pacific Regional Meeting 17th Oct 2023, Merusaka Hotel, Nusa Dua, Bali, Indonesia <https://aprm2023.iatca.or.id/>

Keynote Speaker, “*Intelligent Transportation Systems for Sustainable Smart Cities*”, 5th International Conference on Future Smart Cities, Xiamen University Malaysia Campus, 18th Nov 2022, Kuala Lumpur, Malaysia (Online) <https://www.ierek.com/events/future-smart-cities-fsc-5th-edition#introduction>

Invited Presentations

Invited Speaker

Speaker, “*Immersive Digital Twins and AI-Augmented Interfaces: Enabling Collaborative and Experiential Learning through Mixed Reality Platforms*” AI for Education Singapore 2025 Conference, NTU Singapore, 18th Nov 2025, Singapore

Speaker, “*Advancing Air Traffic Management with AI and Automation*”, Breakfast Talk, The French Chamber of Commerce in Singapore, 18th Nov 2025, Singapore

Speaker, “*DeepFlow - New method for air traffic flow and capacity management*” European Union-ASEAN Research Partnerships: Tackling Global Challenges Together, European Union mission to ASEAN, National Research and Innovation Agency Office, 18th June 2025, Jakarta, Indonesia

Speaker, “*Flow-centric futures – redefining ATFM with AI*”, SESAR Walking Tour, Airspace World 2025, Feira Internacional de Lisboa, 14th May 2025, Lisbon, Portugal

Speaker, “*The Future of Air Traffic Management powered by AI*”, European Union Aviation Safety Agency’s EU-Asia Pacific workshop on Artificial Intelligence in Aviation, NTU Singapore, 21st Nov 2024, Singapore

Speaker and Moderator, “*AI for Air Traffic Management in Singapore*” AI & Motion, Embassy of Italy, Sala Italia, 24th Sept 2024 Singapore

Speaker, “*Machine Learning for Flying Machines*”, Imperial-TUM-NTU Global Fellows Programme, Imperial College London, 10th July 2024. (Online)

Speaker, “*AI Applications in Airspace Design*”, ASI Symposium: Aviation Trends and Development in Air Traffic Management, SUTD, 7th March 2024, M Hotel, Singapore

- Speaker, *"Applications of AI in Air Traffic Management"*, Middle East Regional Monitoring Agency (MIDRMA) RVSM Seminar International Civil Aviation Organization (ICAO), 9th Oct 2023, Civil Aviation Authority, Manama, Bahrain.
- Speaker, *"AI-based Conformal Automation"* EUROCONTROL - FAA Technical Interchange Meeting Human-Systems Integration, 26th – 27th Oct 2023, Eurocontrol Experimental Center, Bretigny-sur-Orge, France
- Speaker, *"AI use cases in Aviation"*, European Union Aviation Safety Agency AI Days, EASA, 16th – 17th May 2023, EASA Headquarters, Cologne, Germany
- Speaker, *"Sustainable Aviation"*, Aerospace Technologies of the Future, Association of Aerospace Industries (Singapore), Kent Ridge Guild House, 30th August 2022, Singapore
- Speaker, *"A Hybrid AI-Human Air Traffic Management Paradigm"*, Applied Machine Learning Days Conference, EPFL, 26th – 30th March 2022, Lausanne, Switzerland (*Online*)
- Invited Presentation to Deputy Prime Minister of Singapore, *"ATMRI's Research in Artificial Intelligence for Singapore Air Traffic"*, Singapore Air Traffic Control Centre, Changi Village, 12th July 2021, Singapore
- Invited Talk, *"Who Moved My Tower? The Digital Transformation in Air Traffic Management"*, Saab-NTU Joint Lab Webinar, NTU Singapore, Singapore, 27th Nov 2020
- Invited Talk, *"How machines can learn and perform higher order cognitive tasks"*, Online R&D Seminar for CAAS staff, Singapore Aviation Academy, 10th July 2020, Singapore
- Invited Talk, *"Application of Artificial Intelligence to ATM"*, Air Navigation Service Leadership and Management Programme, Singapore Aviation Academy, 22nd Nov 2019, Singapore
- Invited Talk, *"Artificial Intelligence for Next Generation Air Traffic Management"*, Air Traffic Management Bureau, Civil Aviation Administration of China, 13th Nov 2019, ATMB, Beijing, China
- Invited Talk: *"Nature Inspired Computation for Airspace Safety Management"*, Birla Institute of Technology Mesra Jaipur Campus, 9th Feb 2017, Jaipur, India,
- Invited Talk, *"Nature Inspired Computation for Airspace Safety Management,"* École Nationale de l'Aviation Civile (ENAC), French Civil Aviation University, 6th Jan 2017, Toulouse, France
- Invited Talk, *"Nature Inspired Computation for Airspace Safety Management"*, Lancaster University Management School, 11th Jan 2016, Lancaster, UK
- Invited Talk, *"Airspace Collision Risk Modelling"*, Aviation Seminar Series, School of Aviation, University of New South Wales (UNSW), 24th May 2016, Sydney, Australia
- Invited Talk, *"Nature Inspired Solutions for Air Transportation Problems"* Aviation Seminar Series, School of Aviation, University of New South Wales (UNSW), 7th Oct 2014, Sydney, Australia
- Speaker, *"Reduced Vertical Separation Minima System"*, MID RVSM Safety Assessment Seminar, ICAO Middle East Regional Monitoring Board Meeting, 9th – 12th Mar 2014, Swissotel, Manama, Bahrain
- Tutorial Talk, *"Evolutionary Computation in Air Traffic Management Research"*, 6th International Conference on Research in Air Transportation, 27th May 2014, Istanbul, Technical University, Istanbul, Turkey
- Invited Talk, *"Computational Red Teaming for Air Traffic Management"*, Seminar Series, Inst. for Integrated & Intelligent Systems, Griffith University, Brisbane, Australia, 10th Mar 2014

Panel Speaker

- Panel Speaker, *"AI research for Air Traffic in Asia Pacific"*, 1st Air Transportation Research and Development Symposium, 26th June 2025, Prague Airport Convention Centre, Prague, Czech Republic

Panel Moderator, “*Should autonomous systems manage and control ATM/UTM?*”, France-Singapore Merlion Workshop on Artificial Intelligence for Air Traffic Management, NTU Singapore, 3rd Oct 2023, Singapore

Panel Moderator, “*Air & Space Traffic Integration*”, Workshop between DLR Germany, EDB Singapore and NTU Singapore, NTU Singapore, 21st Nov 2023, Singapore

Panel Speaker, “*Human-Machine Teaming and Evolution: What is Next?*”, Zero Pressure Podcast by Imperial College London and Saab AB Sweden, 28th Jan 2022 (Online)

Panel Speaker “*A Hybrid AI-Human ATM System*”, SESAR JU (European Union), Fly AI Webinar (Online), 23rd Mar 2021

Panel Speaker, “*A Hybrid AI-Human ATM System*”, CANSO Webinar - Advancing ATM Research & Development in the Asia Pacific, (Online), 1st Oct 2020

Panel Speaker, “*Research Challenges in Air Transport & Digitalisation Challenges*”, Association for Scientific Development of Air Traffic Management in Europe (ASDA), Salzburg Congress Centre, Salzburg, Austria, 3rd Dec 2018

Publications

Bold	Denotes main author
Underline	Denotes 1st academic author
^	Denotes corresponding author
~	Denotes PI/ Supervisor/Team Lead
**	Denotes directly supervised research staff, i.e. POs, RAs, RFs, postdocs, etc.
*	Denotes PhD or research students (supervised or co-supervised)
+	Denotes other students and research staff
##	Denotes Tier 1A papers
#	Denotes Tier 1B papers

Journal Papers

Hu, X.^{**}, Pang, B., **Alam, S.**[~], & Feroskhan, M. (2026). *U-Aerodrome: Data-driven and risk-bounded airspace reconfiguration for safe integration of urban air mobility at aerodrome*. Transportation Research Part C: Emerging Technologies, 184, 105506. (IF: 7.9) ##

Ma, C.^{**}, Mookherjee, D., Mesquida-Masana, G., Dalmau, R., & **Alam, S.**^{^~} (2026). *Enabling interaction-free continuous descent through eligible flight identification and top-of-descent prediction*. Journal of Aircraft. Advance online publication <https://doi.org/10.2514/1.C038399> (IF: 2.2) #

Ali, H.^{**}, **Alam, S.**[~], Donmez, K., & Lun, W. (2026). Machine learning algorithms and models for airport gate assignment problem: A systematic literature review. Transportation Research Part E: Logistics and Transportation Review. Advance online publication. ##

Chen, Y. ^{**}, Dalmau, R., & **Alam, S.**^{^~} (2025). Tactical demand and capacity balancing using incremental search in spatio-temporal graphs with flight uncertainty. Transportation Research Part C: Emerging Technologies, 181, 105382. (IF: 7.9) ##

Ali, H.^{**^}, Hoo, X. T., Thai, V.-P., Pham, D.-T., & **Alam, S.**[~] (2025). Spatial-temporal dynamics of airside congestion: A Macroscopic Fundamental Diagram perspective. Journal of Air Transport Management, 128, 102850. (IF: 3.6) #

Alam, S.[^], Itoh, E., Li, M., Pham, D. T., Schultz, M., & Wang, Y. (2025). Artificial intelligence for airspace optimization and air traffic management. Advanced Engineering Informatics, 65, 103579. (IF: 9.9) ##

Dhief, I.^{**^}, Wang, Z., Zhou, W., **Alam, S.**[~], Stahnke, A., Losensky, L., & Rabus, T. (2025). A review of international best practices in integrating space launch activities with air traffic management: Guidance for future space launch in the Singapore region. Acta Astronautica, 226, 728–759. (IF: 3.4) #

Li, Y., Guo, T., Chen, J., Wu, J., Zhang, Y., & Alam, S. (2025). Urban air mobility: A review and challenges. IEEE Intelligent Transportation Systems Magazine, 17(3), 67–87.(IF: 5.293)

- Zhou, W.** , Wang, Z., **Alam, S.**[^], Materne, L. J., Mühlhausen, T., & Delahaye, D. (2025). Evaluating risk-based hazard corridors in air traffic controller decisions during space launch failures. *Journal of Air Transport Management*, 131, 102904. (IF: 3.6) #
- Ma, C. ^{***^}, Guleria, Y., **Alam, S.**[~], & Li, M. Z. (2025). Deep reinforcement learning-based air traffic flow coordination in flow-centric airspace. *Advanced Engineering Informatics*, 65, 103342. (IF: 9.9) ##
- Zhang, Y.**[^], Xu, S., Zhang, L., Jiang, W., Alam, S., & Xue, D. (2025). Short-term multi-step-ahead sector-based traffic flow prediction based on the attention-enhanced graph convolutional LSTM network (AGC-LSTM). *Neural Computing and Applications*, 37(20), 14869–14888. (IF: 4.5)
- Thai, P.** , Ali, H., Hoo, X. T., Pham, D.-T., & **Alam, S.** (2025). Gaussian mixture variational autoencoder for runway configuration prediction in imbalanced data. *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*. (IF:1.1) #
- Ali, H. ^{***^}, Pham, D. T., **Alam, S.**[~], Schultz, M., Li, M. Z., Wang, Y., Itoh, E., & Duong, V. N. (2025). Human–AI hybrids in safety-critical systems: Concept, definition and perspectives from air traffic management. *Advanced Engineering Informatics*, 65, 103256. (IF: 9.9) ##
- Pang, B.** , Hu, X., Zhang, M., **Alam, S.**[^], & Lulli, G. (2025). A multi-aircraft co-operative trajectory planning model under dynamic thunderstorm cells using decentralized deep reinforcement learning. *Advanced Engineering Informatics*, 65, 103157 (IF: 9.9) ##
- Brun, A.[^], Feron, E., **Alam, S.**[~], & Delahaye, D. (2025). Schedule optimization and staff allocation for airport security checkpoints using guided simulated annealing and integer linear programming. *Journal of Air Transport Management*, 124, 102746 (IF: 3.6) #
- Lim, Y. **, **Alam, S.**[^], Lilith, N., Tan, F., & Toon, P. L. (2025). Graph neural networks with spatiotemporal flow features for aircraft taxi-out-time prediction. *Journal of Air Transportation*, 1–22. American Institute of Aeronautics and Astronautics. (IF: 0.0)
- Ali, H. ^{***^}, Pham, D. T., & **Alam, S.**[~] (2024). Toward greener and sustainable airside operations: A deep reinforcement learning approach to pushback rate control for mixed-mode runways. *IEEE Transactions on Intelligent Transportation Systems*. Volume: 25, Issue: 11 (IF: 8.4) ##+
- Zeng, Y.[^], Hu, M., Chen, H., Yuan, L., Alam, S., & Xue, D. (2024). Improved air traffic flow prediction in terminal areas using a multimodal spatial-temporal network for weather-aware (MST-WA) model. *Advanced Engineering Informatics*, 62, 102935 (IF: 9.9) ##
- Jianzhong, Y.[^], Haoran, H., Yanjun, W., Xiaozhen, M., Minghua, H., & Alam, S. (2024). Robust pre-departure scheduling for a nation-wide air traffic flow management. *Chinese Journal of Aeronautics*. (IF: 5.7) #
- Andy, L. J. G.* , **Alam, S.**[^], Lilith, N., & Piplani, R. (2024). A deep reinforcement learning approach for runway configuration management: A case study for Philadelphia International Airport. *Journal of Air Transport Management*, 120, 102672 (IF: 3.6) #
- Ma, C. **, **Alam, S.**[^], Cai, Q. and Delahaye, D. (2024) Text-Enriched Air Traffic Flow Modeling and Prediction using Transformers, *IEEE Transactions in Intelligent Transportation Systems*, Vol 25, Issue: 7. (IF: 8.4) ##+
- Thai, P.* , **Alam, S.**[^], & Lilith, N. (2024). Airside surveillance by computer vision in low-visibility and low-fidelity environment. *Journal of Air Transportation*, 32(4), 210-217. (IF: 0.0)
- Guleria, Y.* , Pham, D. T., **Alam, S.**[^], Tran, P. N., & Durand, N. (2024). Towards conformal automation in air traffic control: Learning conflict resolution strategies through behavior cloning. *Advanced Engineering Informatics*, 59, 102273. (IF: 9.9) ##
- Chen, K. **, Nadirsha, T. N., Lilith, N, **Alam, S.**[^], and Svensson, A., (2024) Tangible Digital Twin with Shared Visualization for Collaborative Air Traffic Management Operations, *Transportation Research Part C: Emerging Technologies* Volume 161, 104546. (IF: 7.9) ##
- Ferdaus, M. M. ^{***^}, Dam, T., **Alam, S.**[~], & Pham, D. T. (2024). X-Fuzz: An Evolving and Interpretable Neurofuzzy Learner for Data Streams. *IEEE Transactions on Artificial Intelligence*. doi: 10.1109/TAI.2024.3363116. (IF:0.0) #
- Lim, Y.** , **Alam, S.**[^], Tan, F., Lilith, N., (2024). Variable Taxi-Out Time Prediction Based on Machine Learning with Interpretable Attributes, *Transactions of the Japan Society for Aeronautical and Space Sciences* (IF: 0.8)

- Juntama, P.**^, **Alam, S.**, & Delahaye, D. (2024). A study of robustness between two strategic 4D trajectory plans. Transactions of the Japan Society for Aeronautical and Space Sciences, 67(3), 175–183. (IF: 0.8)
- Wang, Z.**^, Delahaye, D., Farges, J. L., & **Alam, S.**~ (2023). A quasi-dynamic air traffic assignment model for mitigating air traffic complexity and congestion for high-density UAM operations. Transportation Research Part C: Emerging Technologies, 154, 104279. (IF: 7.9) ##
- Pham, D. T.**^, Guleria, Y., **Alam, S.**~, & Duong, V. (2023). Probabilistic Conflict Detection using Heteroscedastic Gaussian Process and Bayesian Optimization. IEEE Access, vol. 11, pp. 109341-109352 (IF: 3.6)
- Goh, S.K.***, Lim, Z.J., Singh, N.P.***, and **Alam, S.**^~ (2022), Interpretable Tracking and Detection of Unstable Approaches Using Tunnel Gaussian Process, IEEE Transactions on Aerospace & Electronics Systems, Online ISSN: 1557-9603, Digital Object Identifier: 10.1109/TAES.2022.3217994. (IF: 5.7) ##
- Ali, H.*, Pham^, D.T., **Alam, S.**~ and Schultz, M. (2022), A Deep Reinforcement Learning Approach for Airport Departure Metering under Spatial-Temporal Airside Interactions (2022) IEEE Transactions in Intelligent Transportation Systems, Online ISSN: 1558-0016 (IF: 8.4) ##+
- Cai, Q.***, Ang, H.J., and **Alam, S.**^~, (2022) Collision Risk Assessment of Reduced Aircraft Separation Minima in Procedural Airspaces Using Advanced Communication and Navigation, Chinese Journal of Aeronautics, CJOA-D-22-00261, (IF: 5.7) #
- Lim, Z.J.*., **Alam, S.**^~, Dhief, I., & Schultz, M. (2022), Towards a Greener Extended-Arrival Manager in Air Traffic Control: A Heuristic Approach for Dynamic Speed Control using Machine-Learned Delay Prediction Model, Journal of Air Transport Management, Elsevier, 103 (2022) 102250. (IF: 3.6) #
- Dhief, I.**^, **Alam, S.**~, Lilith, N. & Mean, C.C., A Machine Learned Go-Around Prediction Model using Pilot-in-the-loop Simulations (2022), Transportation Research Part C: Emerging Technologies, Volume 140, July 2022, 103704, (IF: 7.9) ##
- Juntama, P.*^, Delahaye, D., Chaimatanan, S., & **Alam, S.**~ (2022). Hyperheuristic Approach Based on Reinforcement Learning for Air Traffic Complexity Mitigation. Journal of Aerospace Information Systems, 19(9), 633-648) (IF: 1.5)
- Wang, Z.^, Delahaye, D., Farges, J. L., & Alam, S. (2022). Complexity optimal air traffic assignment in multi-layer transport network for Urban Air Mobility operations. Transportation Research Part C: Emerging Technologies, 142, 103776. (IF: 7.9) ##
- Zhong, J. Y.**^, Goh, S. K., Woo, C. J., & **Alam, S.**~ (2022). Impact of Spatial Orientation Ability on Air Traffic Conflict Detection in a Simulated Free Route Airspace Environment. Frontiers in Human Neuroscience, 16. (IF: 2.7) #
- Pham, D.***, Tran, P. N., **Alam, S.**^~, Duong, V. & Delahaye, D. (2022). Deep reinforcement learning based path stretch vector resolution in dense traffic with uncertainties. Transportation Research Part C: Emerging Technologies, 135, 103463. (IF: 7.9) ##
- Tran, P. N.***, Nguyen, H. Q., Pham, D. T., & **Alam, S.**^~. (2022). Aircraft Trajectory Prediction with Enriched Intent Using Encoder-Decoder Architecture. IEEE Access, 10, 17881-17896. (IF: 3.6)
- Ma, C.***, **Alam, S.**^~, Cai, Q., & Delahaye, D. (2022). Critical Links Detection in Spatial-Temporal Airway Networks Using Complex Network Theories. IEEE Access, 10, 27925-27944. (IF: 3.6)
- Thai, P.*, **Alam, S.**^~, Lilith, N., & Nguyen, B. T. (2022). A computer vision framework using Convolutional Neural Networks for airport-airside surveillance. Transportation Research Part C: Emerging Technologies, 137, 103590. (IF: 7.9) ##
- Pham, D.**^, Tran, T., **Alam, S.**~ & Duong, V. N. (2021). A generative adversarial imitation learning approach for realistic aircraft taxi-speed modelling. IEEE Transactions on Intelligent Transportation Systems. 23(3), 2509-2522. (IF: 8.4) ##+
- Pushparaj, K.**^, Ky, G., Ayeni, A. J., **Alam, S.**~ & Duong, V. N. (2021). A quantum-inspired model for human-automation trust in air traffic controllers derived from functional Magnetic Resonance Imaging and correlated with behavioural indicators. Journal of Air Transport Management, 97, 102143. (IF: 3.6) #
- Goh, S. K.***, Lim, Z. J., **Alam, S.**^~ & Singh, N. P. (2021). A tunnel Gaussian process model for learning interpretable flight's landing parameters. Journal of Guidance, Control, and Dynamics, 44(12), 2263-2275. (IF: 2.8) #

- Long, C. L.***, Guleria, Y., & **Alam, S.**[^] (2021). Air passenger forecasting using Neural Granger causal Google trend queries. *Journal of Air Transport Management*, 95, 102083. (IF: 3.6) #
- Cheung, W. L.**[^], Piplani, R., Alam, S., & Bernard-Peyre, L. (2021). Dynamic capacity and variable runway configurations in airport slot allocation. *Computers & Industrial Engineering*, 159, 107480. (IF: 6.5) #
- Schultz, M.**[^], Reitmann, S., & Alam, S. (2021). Predictive classification and understanding of weather impact on airport performance through machine learning. *Transportation Research Part C: Emerging Technologies*, 131, 103119. (IF: 7.9) ##
- Wang, Z.*[^], **Delahaye, D.**, Farges, J. L., & Alam, S. (2021). Air Traffic Assignment for Intensive Urban Air Mobility Operations. *Journal of Aerospace Information Systems*, 18(11), 860-875. (IF: 1.5)
- Zhao, W.**[^], Li, L., Alam, S., & Wang, Y. (2021). An incremental clustering method for anomaly detection in flight data. *Transportation Research Part C: Emerging Technologies*, 132, 103406. (IF: 7.9) ##
- Cai, Q.***, **Alam, S.**[^] and Duong, V.N. (2020). A Spatial-Temporal Network Perspective for the Propagation Dynamics of Air Traffic Delays, *Engineering*, 7(4), 452-464. (IF: 11.6) ##
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- Van Phat*, T., **Alam, S.** , Lilith, N., Tran, P. N., & Nguyen, B. T. (2020) Aircraft Push-back Prediction and Turnaround Monitoring by Vision-based Object Detection and Activity Identification. 10th SESAR Innovation Days (SID), 7-10 Dec, on-line.
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- Ky*, G., **Alam, S.** , & Duong, V.N., (2020) Carousel Inspired Virtual Circulation: A Simulation Model for UAV Arrival and Landing Procedure Under Random Events, WSC 2020, Proceedings of the 2020 Winter Simulation Conference, Florida, USA, 23-26 June 2020
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- Pham, D.T.**., Ngo, M., Tran, N., **Alam, S.**, Duong, V. (2019). A Data-Driven Approach for Taxi-Time Prediction: A Case Study of Singapore Changi Airport. ENRI International Workshop on ATM/CNS (EIWAC 2019), Tokyo, Japan
- Schultz, M.** and Reitmann, S. and Alam, S. (2019), Classification of Weather Impacts on Airport Operations, 2019 Winter Simulation Conference (WSC), Washington, USA
- Piplani, R.**, Cheung, W.L., Alam, S. and Bernard-Peyre, L., (2019) Airport schedule optimization to address peak period demand capacity imbalance, ENRI International Workshop on ATM/CNS (EIWAC2019), Tokyo, Japan
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- Alam, S.**, Chaimatanan, S., Delahaye, D., and Féron, E. (2017) A Distributed Air Traffic Flow Management Model for European Functional Airspace Blocks, 12th ATM R&D Seminar, Seattle, USA
- Hossain*, M., and **Alam, S.** (2016). Intelligent anticipatory agents for changing environments. In Systems, Man, and Cybernetics (SMC), 2016 IEEE International Conference on Systems, Man, and Cybernetics, SMC, 2016, (pp. 004597-004602). Budapest, Hungary
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- Symon**, F., **Alam, S.**, Blom H. (2014), Airspace Network Characterization for Effect of Intermediate Waypoints on Collision Risk Assessment, 6th International Conference on Research in Air Transportation, Istanbul, Turkey
- El-Sergany*, A. T., Griffin, A., Tranter, P., **Alam, S.** (2014), Descriptive and Geographical Analysis of Flood Disasters Evacuation Modelling, 11th International Conference on Information Systems for Crisis Response and Management (ISCRAM 2014), Pennsylvania, USA
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- Alam, S.**, Ngugen, M.H., Abbass, H.A., Ellejmi, M. and Kirby, S.(2010), A Dynamic Continuous Descent Approach Methodology for Low Noise and Emission, 29th IEEE/AIAA Digital Avionics Systems Conference (DASC), Salt Lake City, UT, USA
- Alam, S.**, Lokan C.J., Abbass, H.A., Ellejmi, M. and Kirby, S. (2010) An Evolutionary Computational Analysis of Tactical Controller Tool, In Proc. Aviation Information Technology Engineering and Management Conference (AITEM2010), New Orleans, Louisiana, USA
- Purton*, L., Abbass, H.A., **Alam, S.** (2010) Identification of ADS-B System Vulnerabilities and Threats, 33rd Australian Transport Research Forum, Canberra, Australia
- Zhao, W.**, Tang, J., Alam, S., Bender, A., Abbass, H. (2010) Evolutionary-Computation Based Risk Assessment of Aircraft Landing Sequencing Algorithms, In Proc. 3rd IFIP Conference on Biologically Inspired Collaborative Computing, Brisbane, Australia
- Alam, S.**, Abbass, H.A., Lokan C.J. (2009) Computational Red Teaming to Investigate Failure Patterns in Medium Term Conflict Detection, 8th Eurocontrol Innovation Research Workshop and Conference, Eurocontrol Experiment Research Centre, Paris, France
- Tang, J.**, Alam, S., Abbass, H.A. and Lokan, C.J. (2009) Modelling and Evolutionary Multi-objective Evaluation of Interdependencies and Work Processes in Airport Operations, World Summit Asia-Pacific Conference on Simulated Evolution on Genetic and Evolutionary Computation, ACM 978-1-60558-326, Shanghai, China
- Ngugen, M. H.**, Alam, S., Abbass, H.A. (2007) Dynamic Weather Avoidance in a Traffic Constrained Airspace, 6th Eurocontrol Innovation Research Workshop and Conference, Eurocontrol Experiment Research Center, Paris, France

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Alam, S. , Ngugen, M.H., Abbass, H.A. and Barlow, M. (2007) On Architecture, Design and Validation of Air Traffic Simulator, SIMTECT-07, Brisbane, Australia

Alam, S., Bui, L.T., Abbass, H.A., Barlow, M. (2006). Pareto Meta-heuristics for Generating Safe Flight Trajectories Under Weather Hazards, Asia-Pacific Conference on Simulated Evolution and Learning (SEAL 2006), Hefei, China.

Alam S. , Abbass H.A., Barlow M. and Lindsay P. (2005) Mapping Lessons from Ants to Free Flight: An Ant-based Weather Avoidance Algorithm in Free Flight Airspace, SPIE05 Complex Systems, vol. 6039, Brisbane, Australia

Innovation, Intellectual Property & Consulting

Patents Filed

S/No.	Patent No.	School/ Center	Title	NTU Inventors	Status
1	2019-10201912771V	ATMRI	Aircraft Trajectory Modelling and Unstable Approach Detection In Terminal Manoeuvring Area (TMA)	GOH Sim Kuan (MY); Narendra Pratap SINGH (IN); Sameer ALAM (AU)	Accept - SG PRV 10201912771V

Technology Disclosure Filed

S/No.	TD No.	School/ Center	Title	NTU Inventors	Status
1	2025-326	ATMRI	Exploring Airlines' Scheduled Buffer Time Adjustment Strategies: An Analytical Approach	Zhou Ying; Sameer Alam	Technology Disclosure
2	2025-242	ATMRI	Optimized Sequencing and Conflict-Free Path Planning for Arrival Flights during Runway Direction Change	Jiang Hao; Zhi Jun Lim; Sameer Alam	Technology Disclosure
3	2025-230	ATMRI	Automating Terminal Airspace Vectoring: A Machine-Assisted Approach for Sequencing, Spacing, and Merging Arrival Flights	Imen Dhief; Zhi Jun Lim; Sameer Alam	Technology Disclosure
4	2025-229	ATMRI	A Probabilistic Framework for Real-Time Processing of Aircraft Surface Movement Data in Digital Twins	Nam Tran; Duc-Thinh Pham; Sameer Alam	Technology Disclosure
5	2025-228	ATMRI	Text2Traffic: Retrieval-Enhanced In-Context Learning for Complex Air Traffic Scenario Generation	Yash Guleria; Sameer Alam; Pham Duc Thinh; Ashton Low Kin Yun; Mohammed Nadirsha	Technology Disclosure
6	2025-227	ATMRI	A Data-Driven Framework for Modelling Complexity in Terminal Manoeuvring Area	Lim Zhi Jun; Sameer Alam; Imen Dhief; Pham Duc Thinh	Technology Disclosure
7	2025-226	ATMRI	Tactical Demand and Capacity Balancing with Uncertainty Using Incremental Path-Search Based on Spatio-Temporal Graph	Yutong Chen; Sameer Alam	Technology Disclosure
8	2025-225	ATMRI	Risk-Aware Flight Trajectory Planning Under Convective Weather	Wei Zhou; Yutong Chen; Sameer Alam	Technology Disclosure
9	2025-224	ATMRI	Speech-to-Route: Leveraging Large Language Models for Taxi Route Visualization	Thai Van Phat; Sameer Alam	Technology Disclosure

S/No.	TD No.	School/ Center	Title	NTU Inventors	Status
10	2025-223	ATMRI	ATCO-Centered Explainable AI	Katherine Fennedy; Sameer Alam; Mohammed Nadirsha; Li Hua	Technology Disclosure
11	2024-387	MAE	Dynamic Air Traffic Flow Coordination for Flow-Centric Airspace Management	Chunyao Ma; Sameer Alam; Qing Cai	Technology Disclosure
12	2024-298	MAE	A Monte Carlo Simulation Framework with Agent-Based Modeling for Space Launch Impact and Risk Assessment on Air Traffic	Wei Zhou; Zhengyi Wang; Sameer Alam	Technology Disclosure
13	2024-289	MAE	A Reinforcement Learning Approach for Flight Rerouting in Space Launch Failure Scenarios with Dynamic Debris Hazards	Wei Zhou; Zhengyi Wang; Sameer Alam	Technology Disclosure
14	2024-128	MAE	Integrated Air and Space Traffic Management: An Agent-Based Simulation for Analysis of Space- Launch Impact on Air Traffic	Wei Zhou; Zhengyi Wang; Sameer Alam	Technology Disclosure
15	2023-377	MAE	Collaborative Mixed Reality Digital Twin for Airport Tower Management	Mohammed Nadirsha; Sameer Alam; Nimrod Lilith; Pallavi Mohan; Chen Ken	Accept – Patent Filing Consideration
16	2023-372	ATMRI	An AI-Enabled Air Traffic Conflict Detection and Resolution Software for Non-Nominal Traffic Scenarios	Pham Duc Thinh; Yash Guleria; Mohamed Arif bin Mohamed; Dang Huu Phuoc; Sameer Alam	Submission – Know-How
17	2023-256	ATMRI	A Machine Learning Framework for Balancing Air Traffic Demand and Airspace Capacity through Dynamic Airspace Sectorization	Wei Zhou; Duc-Thinh Pham; Sameer Alam	Submission – Know-How
18	2023-250	ATMRI	A Runway Exit Prediction Model with Visually Explainable Machine Decisions	Sameer Alam; Goh Sim Kuan; Woo Chuan Jie; Md Meftahul Ferdous	Submission – Copyrighted Software
19	2023-242	ATMRI	A Multi-View Vision-Based Deep Learning Model for Aircraft Distance- to-Touchdown Prediction	Pham Duc Thinh; Gabriel James Goenawan; Sameer Alam	Submission – Know-How
20	2022-308	ATMRI	A Deep Reinforcement Learning Approach for Airport Departure Metering	Hasnain Ali; Pham Duc Thinh; Sameer Alam	Accept – Copyrighted Software
21	2022-182	MAE	Machine Learning Taxi Time Prediction for Air Traffic Control	Lim Yi Xiang; Toon Pei Ling; Sameer Alam; Nimrod Lilith	Accept – Copyright
22	2022-163	MAE	Machine Learning Go-Around Prediction Assistant for Air Traffic Control	Imen Dhief; Sameer Alam; Nimrod Lilith	Accept – Copyright
23	2021-112	ATMRI	A Data-Driven Approach for Taxi- Route and Taxi-Time Prediction	Pham Duc Thinh; Tran Thanh Nam; Sameer Alam; Vu Nguyen Duong	Accept – Copyright
24	2021-111	ATMRI	A Generative Adversarial Imitation Learning Approach for Aircraft Ground Movement Model	Pham Duc Thinh; Tran Thanh Nam; Sameer Alam; Vu Nguyen	Accept – Copyright

S/No.	TD No.	School/ Center	Title	NTU Inventors	Status
				Duong	
25	2021-087	ATMRI	Aircraft Trajectory Prediction with Enriched Intent Using Encoder–Decoder Architecture	Tran Ngoc Phu; Nguyen Quang Viet Hoang; Pham Duc Thinh; Sameer Alam	Accept – Copyright
26	2021-056	MAE	ConvNet-Based Computer Vision for Airport Airside Monitoring	Sameer Alam; Nimrod Lilith; Tran Ngoc Phu	Accept – Copyright
27	2021-055	MAE	ConvNet-Based Computer Vision for Aircraft Turnaround Time Prediction	Sameer Alam; Nimrod Lilith; Tran Ngoc Phu	Accept – Copyright
28	2019-347	Saab-NTU Corp Lab	Detection, Tracking and Classification of Aircraft and Drones in Digital Towers Using Machine Learning on Motion Patterns	Duong Nguyen-Vu; Sameer Alam; Pham Duc Thinh	Accept – Copyright
29	2019-185	ATMRI	Data-Driven Approaches for Conflict Detection Using Generative Model from Trajectory Data	Pham Duc Thinh; Tran Ngoc Phu; Sameer Alam; Vu Nguyen Duong	Accept – Copyright
30	2019-184	ATMRI	An Intelligent Interactive Conflict Solver Incorporating Air Traffic Controllers' Preferences Using Reinforcement Learning	Tran Ngoc Phu; Pham Duc Thinh; Goh Sim Kuan; Sameer Alam; Vu Nguyen Duong	Accept – Copyright
31	2019-164	ATMRI	Data-Driven Network Approaches for Air Traffic Delay Analysis	Cai Qing; Yash Guleria; Sameer Alam; Ma Chunyao	Accept – Copyright

Consulting & Entrepreneurship

Consulting Income:

Year	Organization	Area of Consultation	Amount (SGD)
2022-2023	Middle East Regional Monitoring Agency of the ICAO	Airspace Collision Risk Modelling	137,000
2021-2023	Saab Singapore	Digital Tower Feasibility for ASEAN region	24,000
2020-2021	Ministry of Transport and Telecommunication, Bahrain	State Action Plan for Aviation Emission	22,000

Teaching, Supervision & Mentoring

Courses Taught

Course Code	Course Title	Academic Year	Course Level
MA6904	Air Transportation System	AY26 onwards	PG (Lec+Tut)
MA7932	Air Transportation Systems	AY21 – AY22	PG (Lec +Tut)
MA4881	Air Traffic Management	AY19 - present	UG (Lec+Tut)
MA0218	Intro to Data Science & Artificial Intl.	AY19 - AY23	UG (Lec)
MA4880	Airport Operations and Systems	AY18 - present	UG (Lec +Tut)
MA6932	Fundamentals of Air Traffic Control	AY18 – AY20	PG (Lec+Tut)
MA1008	Introduction to Computational Thinking	AY18 – AY19	UG (Tut)
MA2006	Engineering Mathematics	AY17 – AY18	UG (Tut)

Academic Supervision & Mentoring

PhD students

No	PhD Student	Period	Role	Thesis/ Project Title	Current Status
Current					
1	Chan Chea Mean	2024 - Present	Sole Supervisor	Human-AI Hybrid Models for Aviation	Ongoing
2	Ying Zhou	2024 - Present	Sole Supervisor	Learning-Based Approach for Air Traffic Management Decision Making	Ongoing
3	Minh H. Le	2024 - Present	Sole Supervisor	Multi-Agent for Automated Inter-Airline Slot Swapping	Ongoing
4	Nguyen Phan Nhat Hoang	2023 - Present	Sole Supervisor	Multi Agent Air Traffic Flow Management	Ongoing
5	Nguyen Duy Anh	2020 - Present	Co-Sup	Reinforcement Learning for Multi-Airport Slots Scheduling with Fairness Consideration	Submitted Thesis
Graduated					
1	Zhang Na	2021 - 2024	Main Supervisor	Collision risk modeling and resolution path planning for UAV conflict management in dynamic environments	RF, EEE, NTU
2	Yash Guleria	2020 - 2024	Sole Supervisor	Data Driven Ground Delay Management under Weather and Traffic Uncertainties	Asst Prof, IIT Mandi, India
3	Lam Jun Guang Andy	2019 – 2025	Main Supervisor	Machine learning for airport management and tower control	Senior Scientist, A*STAR, Singapore
4	Lim Zhi Jun	2019 - 2024	Sole Supervisor	A Multi-Agent Approach for Extended Arrival Manager in Fragmented Airspace using Machine Learning Methods	Lecturer, MAE NTU
5	Thai Van Phat	2019 – 2023	Main Supervisor	Detection, tracking and classification of small flying objects in wide-area	RF, ATMRI, NTU
6	Chun Yao Ma	2018 - 2023	Sole Supervisor	Machine Algorithms for Airspace Safety and Traffic Flow Management	Asst Prof, SIT, Singapore
7	Hasnain Ali	2018 - 2022	Sole Supervisor	Machine Learning Algorithms for Airport Surface Movement Optimization	Data Scientist, Public Service, Australia
8	Ky Gregoire Arthur	2021 – 2025	Co-Supervisor	Virtual Centralized Air Traffic Flow Management	RF, ATMRI, NTU
9	Zeng Yixi	2019 – 2023	Co-Supervisor	Machine learning for hybrid AI-Human Control-Automation	RF, ATMRI, NTU
10	Wai Lun Cheung	2015 - 2021	Co-Supervisor	Optimizing arrival and departure management coupled with surface management and runway optimization techniques	Data Scientist, Thales, Singapore

Teaching Awards & Recognition

2026	Teaching project selected as one of the three AI in Education demonstrations to be showcased to the Minister of Education.
2025	Best Paper Award, 25th Integrated Communications, Navigation and Surveillance Conference, Brussels, Belgium (Chea Mean Chan).
2025	Best Paper Award, 25th Integrated Communications, Navigation and Surveillance Conference, Brussels, Belgium (Nam Thanh Tran).
2025	Best Paper Award, 44th Digital Avionics Systems Conference, Montreal, Canada (Lim Zhi Jun).

2025	Best Paper Award, 44th Digital Avionics Systems Conference, Montreal, Canada (Thaivalappil Noushad).
2024	European Unions' SESAR Young Scientist Award to research staff Dr Yutong Chen. 10,000 Euro Cash Prize
2024	Second position in European Unions' SESAR Young Scientist Award to research staff Dr Wang Zhengyi.
2023	Women in Engineering, Science, and Technology (WiEST) Development Fellowship Grant to PhD student Zhi Jun Lim
2023	Imperial-TUM-NTU Global Fellowship award to PhD student Zhi Jun Lim
2022	Mistletoe Research Fellowship (MRF) award 2022-23 to PhD student Hasnain Ali
2021	Best Student Paper Award at the IEEE Integrated Communications Navigation and Surveillance Conference (ICNS 2021) to PhD student Lam Jun Guang Andy
2021	Singapore Data Science Consortium (SDSC) Dissertation Research Fellowship award to PhD student Hasnain Ali
2023	Mentored 7 UG Engineering Innov. Design students who won 1st Prize (Safety and Health Category) in the 20th Engineering Innovation & Design (EID) Open House and Competition.
2019	Mentored 6 UG Engineering Innov. Design students who won 3 rd Prize in 16th Engineering Innovation & Design (EID) Open House and Competition.
2018	Mentored 7 UG Engineering Innov. Design students who won 1 st Prize in 15th Engineering Innovation & Design (EID) Open House and Competition.

Academic Leadership & Professional Service

School Service

Period of appointment	Role
2025 - Present	Member, School Search Committee
2020 - Present	Member, Accreditation Committee (Aerospace Eng) Programme
2019 - 2024	Deputy Director, Air Traffic Management Research Institute, MAE NTU
2019 - 2024	Co-Director, Saab-NTU Joint Lab, MAE NTU
2018 - 2024	Program Director, Air Traffic Management Research Institute, MAE NTU

University Service

Period of appointment	Role
2025 - Present	Director, Air Traffic Management Research Institute [^] , NTU
2025	Member, AI4X Post Doc Fellowship Evaluation Committee

[^]Air Traffic Management Research Institute (ATMRI) is a Pan-University Research institute effective 2025.

Academic & Professional Community

Period of Appointment	Role
2026	External Reviewer for Research Grants Council (RGC) of Hong Kong 2026
2026	Associate Editor, IEEE Trans on Aerospace & Electronic System
2025	Associate Guest Editor, Journal of Air Transport (Special Issue), 2025
2025	Managing Guest Editor, Advanced Engineering Informatics (Special Issue), 2025
2024, 2025, 2026	Member of the Executive Committee, FAA & Eurocontrol's Air Transportation Research and Development Seminar (ATRDS) 2025
2024, 2025, 2026	Member of the Program Committee, SESAR Innovation Days, (nominated by SESAR JU of European Commission).
2024	Member of the Executive Committee and Organizing Chair, FAA & Eurocontrol's International Conference on Research in Air Transportation 2024
2024	Associate Guest Editor, AIAA Journal of Air Transport (Special Issue)
2023,2024	Associate Editor, IEEE Transactions on Artificial Intelligence
2023	Associate Guest Editor, AIAA Journal of Air Transport (Special Issue)

Period of Appointment	Role
2023	Panel of Judges, Airports Council International (ACI) Asia-Pacific - Young Executive Award 2023
2018 - Present	Editorial Board Member, Elsevier Transportation Research Part C: Emerging Technologies (IF: 9.022)
2022	External Reviewer, Singapore Israel Industrial Research and Development Foundation by Enterprise Singapore, Ministry of Trade, and Industry, Singapore
2021-2024	Visiting Fellow, University of New South Wales, Canberra, Australia
2021	Member of the Program Committee, US & Europe ATM R&D Seminar (nominated by US Federal Aviation Administration)
2021	External reviewer for National Research Foundation (NRF) research proposal (CNRS@CREATE)
2017 - 2018	Managing Guest Editor, Special Issue on Artificial Intelligence in Air Transportation, Elsevier Transportation Research Part C: Emerging Technologies (IF: 9.022)
2015 - 2016	Activity Chair, IEEE Computational Intelligence Society, Canberra Chapter, Australia

External PhD Examination

Period of appointment	Role	University
2025	PhD Thesis Examiner for Mr. Zakaria Ezzahed	National d'Études et de Recherches Aérospatiales (ONERA), Toulouse, France
2025	PhD Thesis Examiner for Ms Ye Liu	Hong Kong Polytechnic University, Hong Kong
2025	PhD Thesis Examiner for Ms Fateme Baneshi	Universidad Carlos III de Madrid, Spain
2024	PhD Thesis Examiner for Mr. Thinh Hoang-Dinh	University of Toulouse, France
2024	PhD Thesis Examiner Alexis BRUN	ECOLE NATIONALE DE L'AVIATION CIVILE, France
2024	PhD Thesis Examiner for Loïc Shi-Garrier	University of Toulouse, France
2021	Habilitation Thesis Examiner for Dr.-Ing. Ms Judith Rosenow	Technische Universität Dresden, Germany
2021	PhD Thesis Examiner for Ms.Margarita Bagamanova	Universitat Autònoma de Barcelona, Catalonia, Spain
2020	PhD Thesis Examiner for Mr. Huang Feng	University of South Florida, Tampa, FL USA
2019	PhD Thesis Examiner for Mr Duc-Thinh Pham	Université Paris Sciences et Lettres, France
2018	PhD Thesis Examiner for Ms Ren Pan	City University of Hong Kong, Hong Kong
2018	PhD Thesis Examiner for Ms Imen Dheif	University of Toulouse, France
2017	PhD Thesis Examiner for Ms Annie Man Liang	University of Toulouse, France